

Thomas F. Freddo, O.D., Ph.D.
Professor
MCP Health Sciences University

Rigid Connective Tissues – Cartilage and Bone

Cartilage

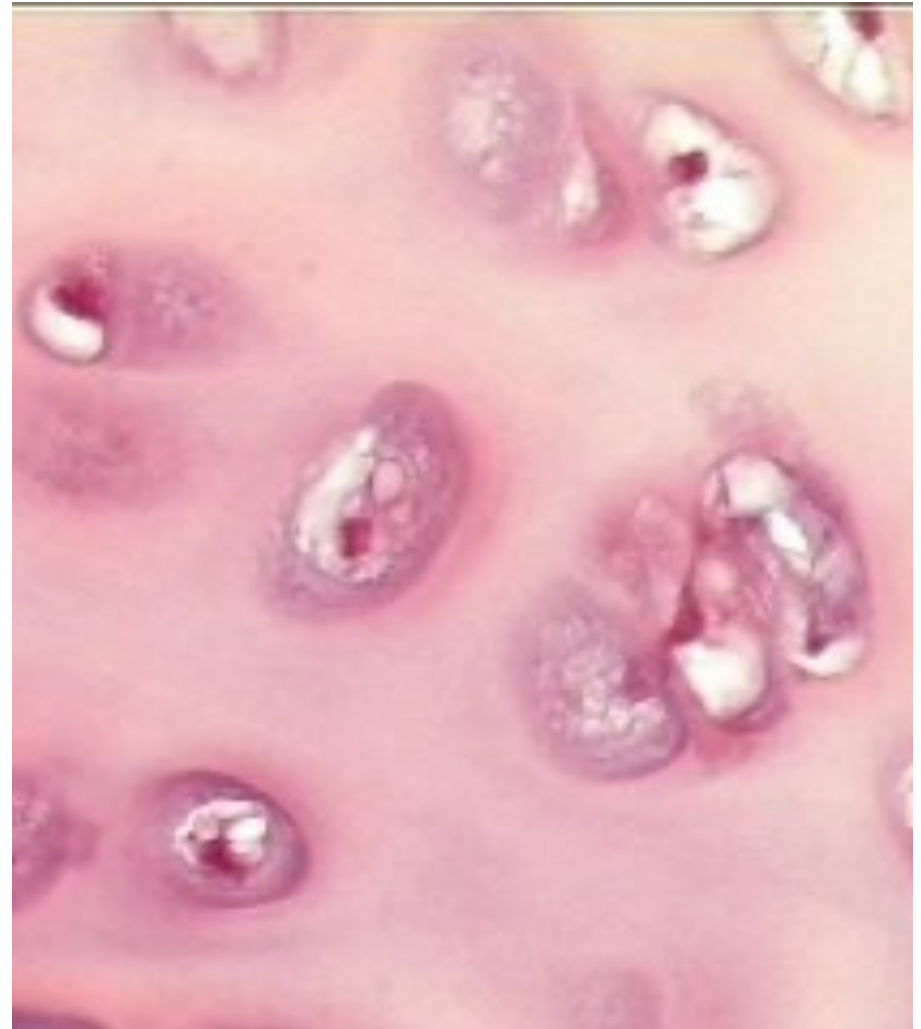
- Forms supporting framework of some organs, such as the walls of airways (nose, trachea, larynx and bronchi), where it prevents airway collapse.
- Forms the articulating surfaces of long bones
- Forms the template for the growth and development of long bones, and most of the rest of the fetal skeleton except the skull (gradually replaced by bone – endochondral ossification). In children, the cartilaginous plates at the ends of long bones can be seen on X-rays. These templates disappear when adults reach their full height.
- It is an AVASCULAR tissue
- Cartilage can (like cornea) generally be transplanted without rejection

Cartilage (prefix –chondro)

- Strong but flexible and semi-rigid supporting tissue. It can withstand tremendous compression forces, and yet it can bend.
- Cellular components - **chondroblasts** and **chondrocytes**,
- Extracellular components: made up about 10% **aggrecan (also found in sclera – it is a form of chondroitin sulfate that aggregates with hyaluronan)**, 75% **water**, and a mix of **collagen fibers** subtypes.
- Three types of cartilage, which differ mostly in the type of fibers they contain.
- **Hyaline**: Most common - has a glassy appearance (hyalos – is Greek for glass).
- **Fibrocartilage**: tendon insertions and intervertebral discs: reinforced with parallel bundles of collagen fibers.
- **Elastic cartilage**: external ear and epiglottis: flexible and resilient - has elastic fibers as well as collagen fibers.

Hyaline Cartilage

- This type of cartilage has a glassy appearance when fresh, hence its name, as *hyalos* is Greek for glassy. It looks slightly basophilic overall in H&E sections.
- **Hyaline cartilage** has widely dispersed fine, Type II **collagen fibers**, which strengthen it.
- It has a **perichondrium**.
- Found covering bones at joints, in ribs, nose, larynx (except epiglottis), and trachea.

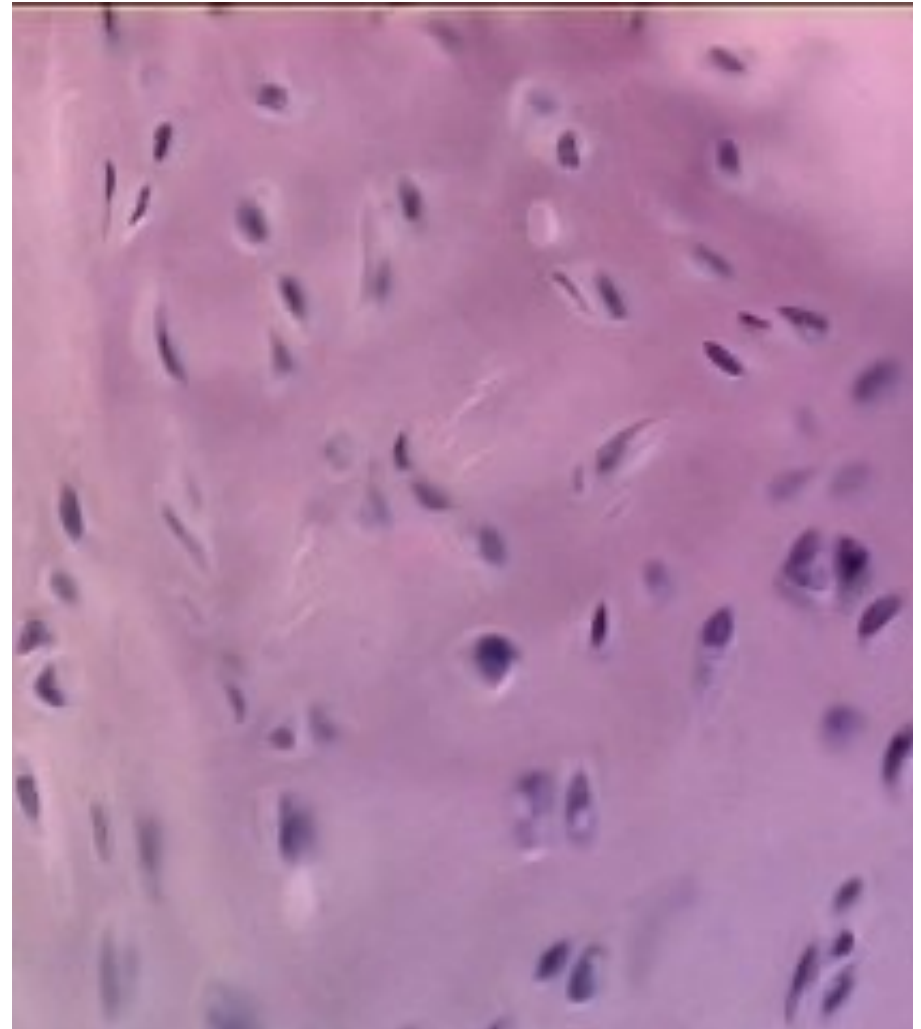


Perichondrium



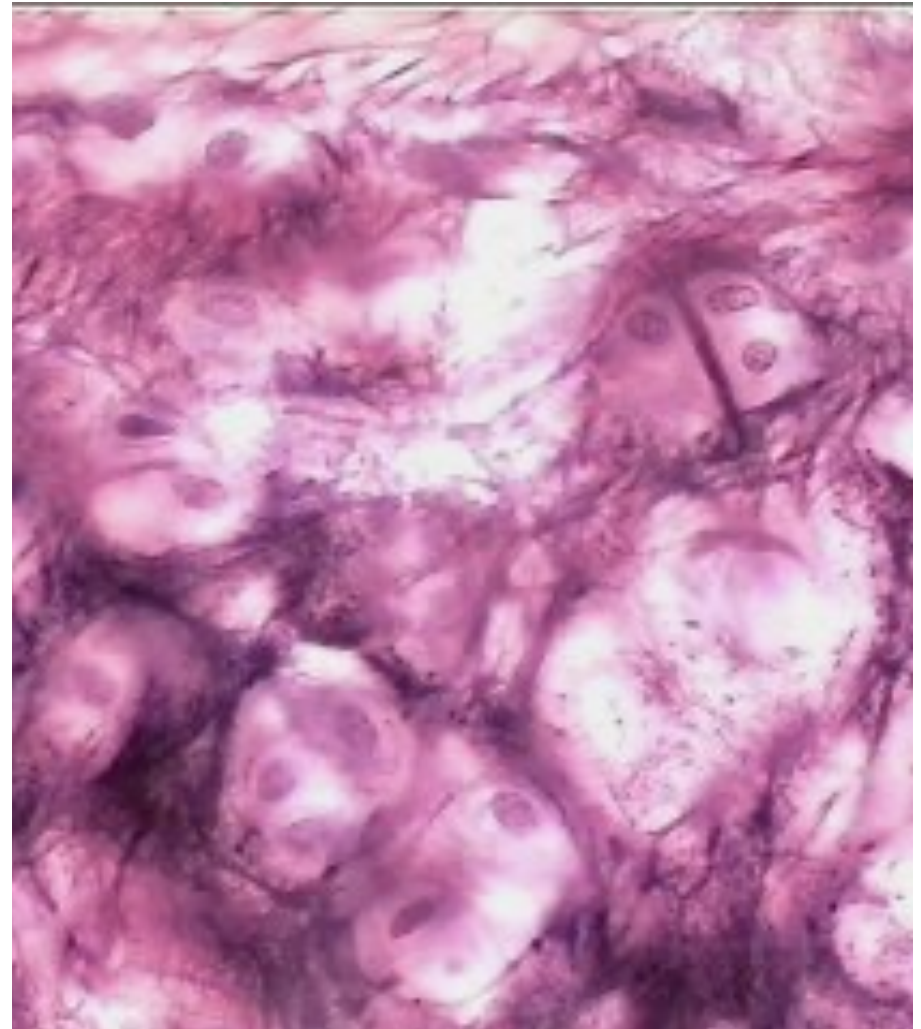
Fibrocartilage

- This is the **strongest** kind of cartilage, because it has alternating layers of hyaline cartilage matrix and thick layers of **dense collagen fibers** oriented in the direction of functional stresses. Collagen is indistinct to matrix and thus invisible at LM level.
- This type of cartilage *does not* have a **perichondrium** as it is usually a **transitional layer** between hyaline cartilage and tendon or ligament.
- Examples include pubic symphysis and intervertebral discs, joint capsules, ligaments.

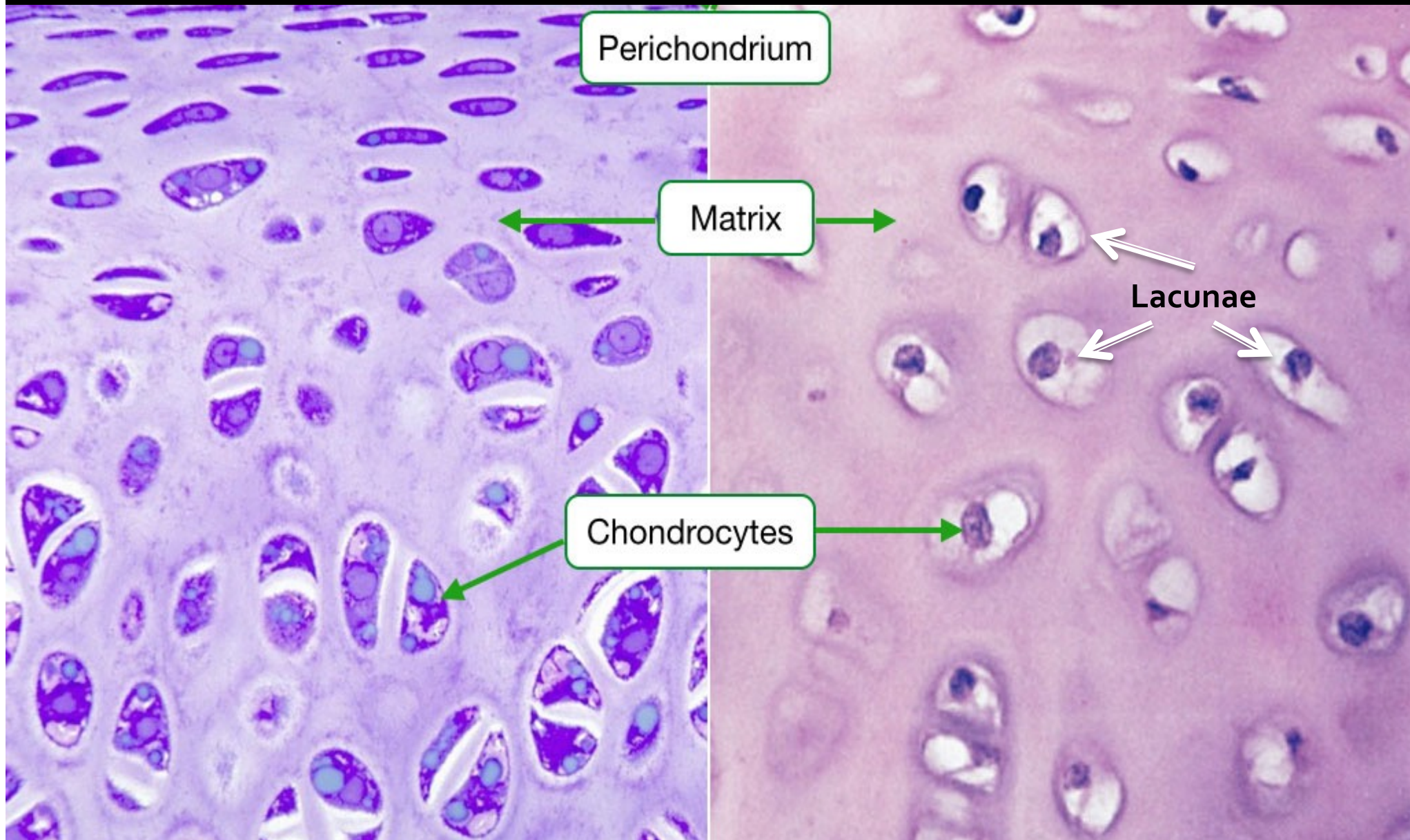


Elastic Cartilage

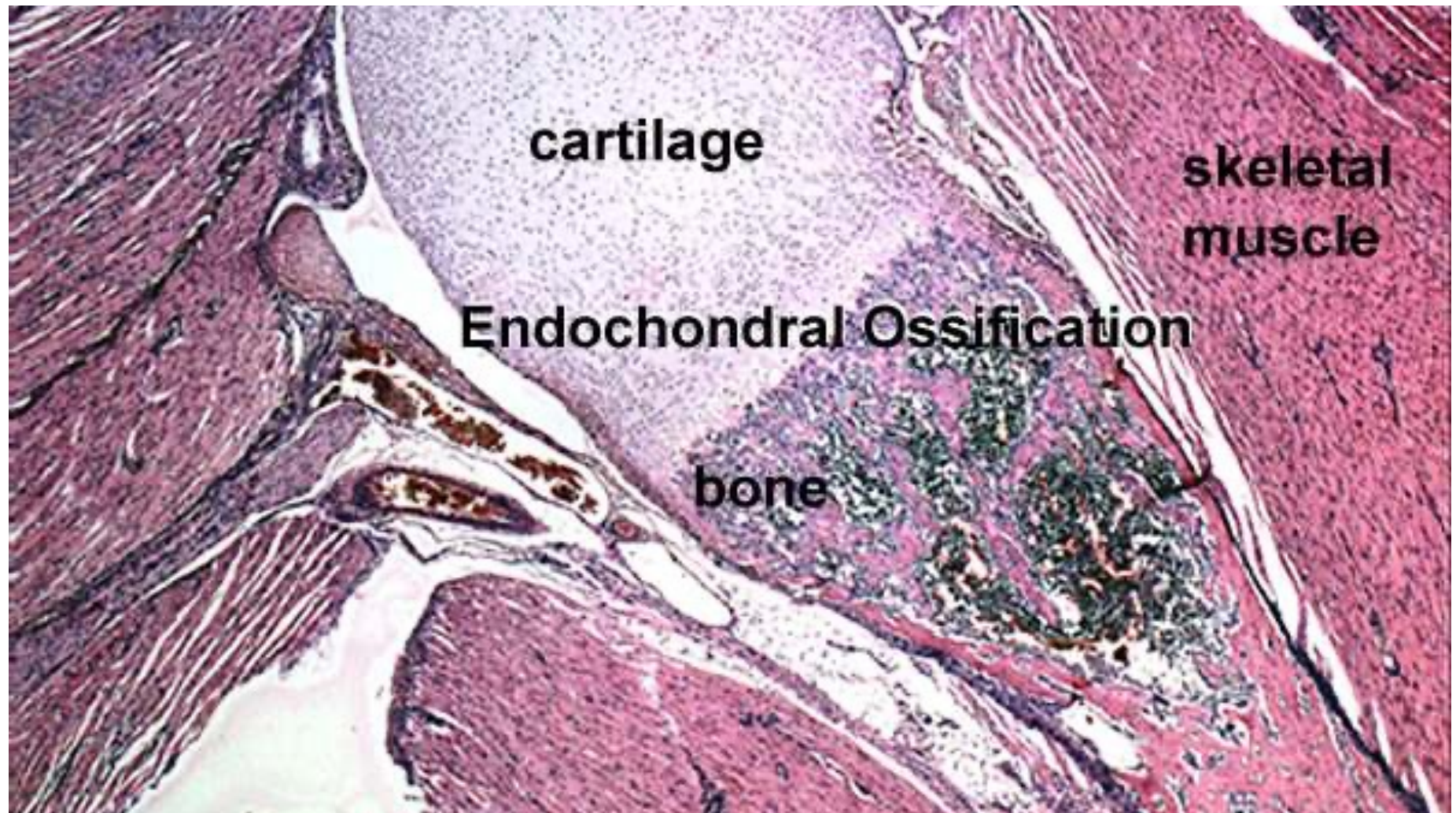
- In **elastic** cartilage, the chondrocytes are found in a threadlike network of elastic fibers.
- **Elastic** cartilage provides strength, and elasticity, and maintains the shape of certain structures such as Eusatachian tubes, epiglottis, and ear lobes.
- It has a **perichondrium**.



Cells of Cartilage – Chondrocytes which when mature are found within lacunae



Cartilage at the articulating tips of long bones



Bone and its Functions

- **Support** - bones make up a structural framework for the body, and provide attachment sites for muscles.
- **Protection** - protection of internal organs - i.e. brain, heart and lungs.
- **Assisting movement.**
- **Mineral homeostasis** - a storage place for calcium and phosphorus
- **Blood cell production** - takes place in the bone marrow.

Bone (prefix – osteo)

- Bone is a strong, flexible and semi-rigid supporting tissue. It can withstand compression forces, and yet it can bend.
- Like cartilage, and other types of connective tissue, bone is made up of:
- **Cells and Extracellular matrix and inorganic compounds (calcium phosphates)**

Overview of Bone Anatomy

- <https://www.youtube.com/watch?v=teq9V7dPjLg>

Two Areas of Bone

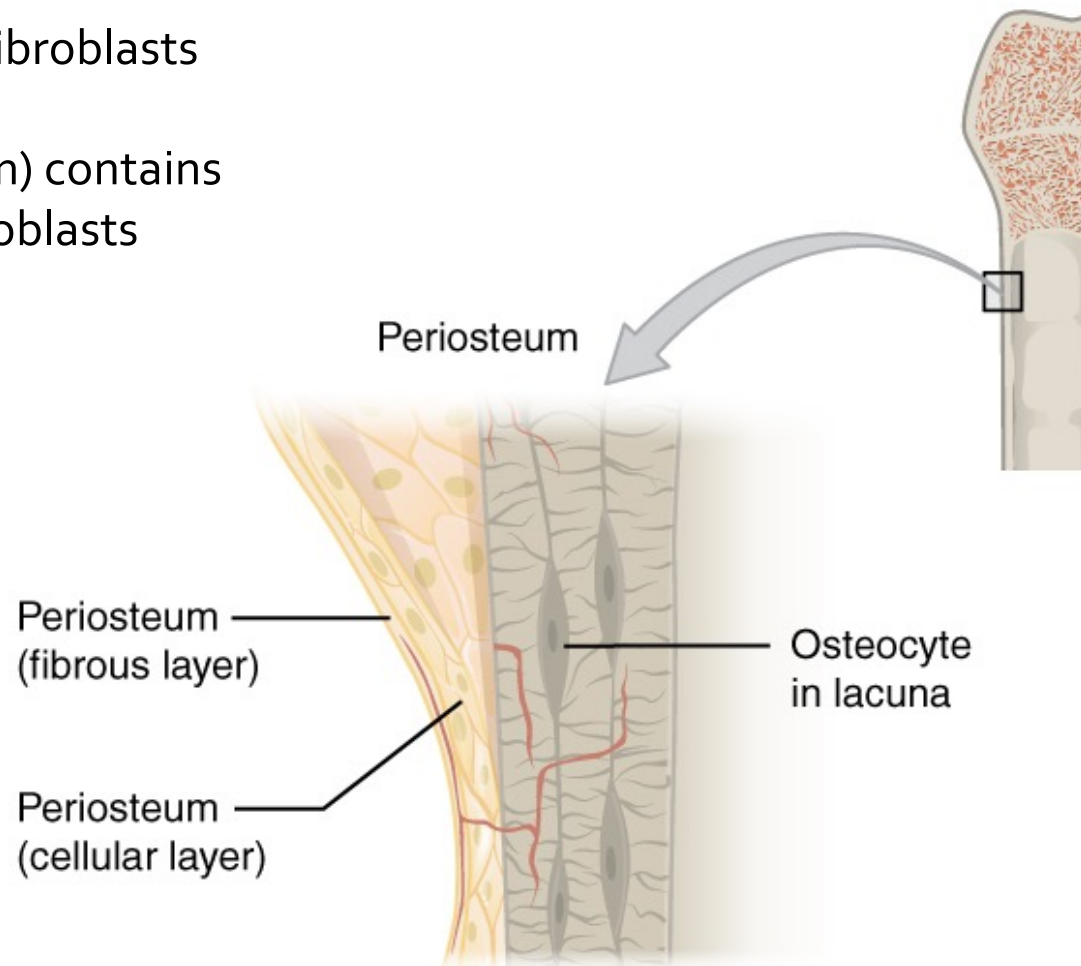
- **Compact Bone**
- **Spongy Bone**



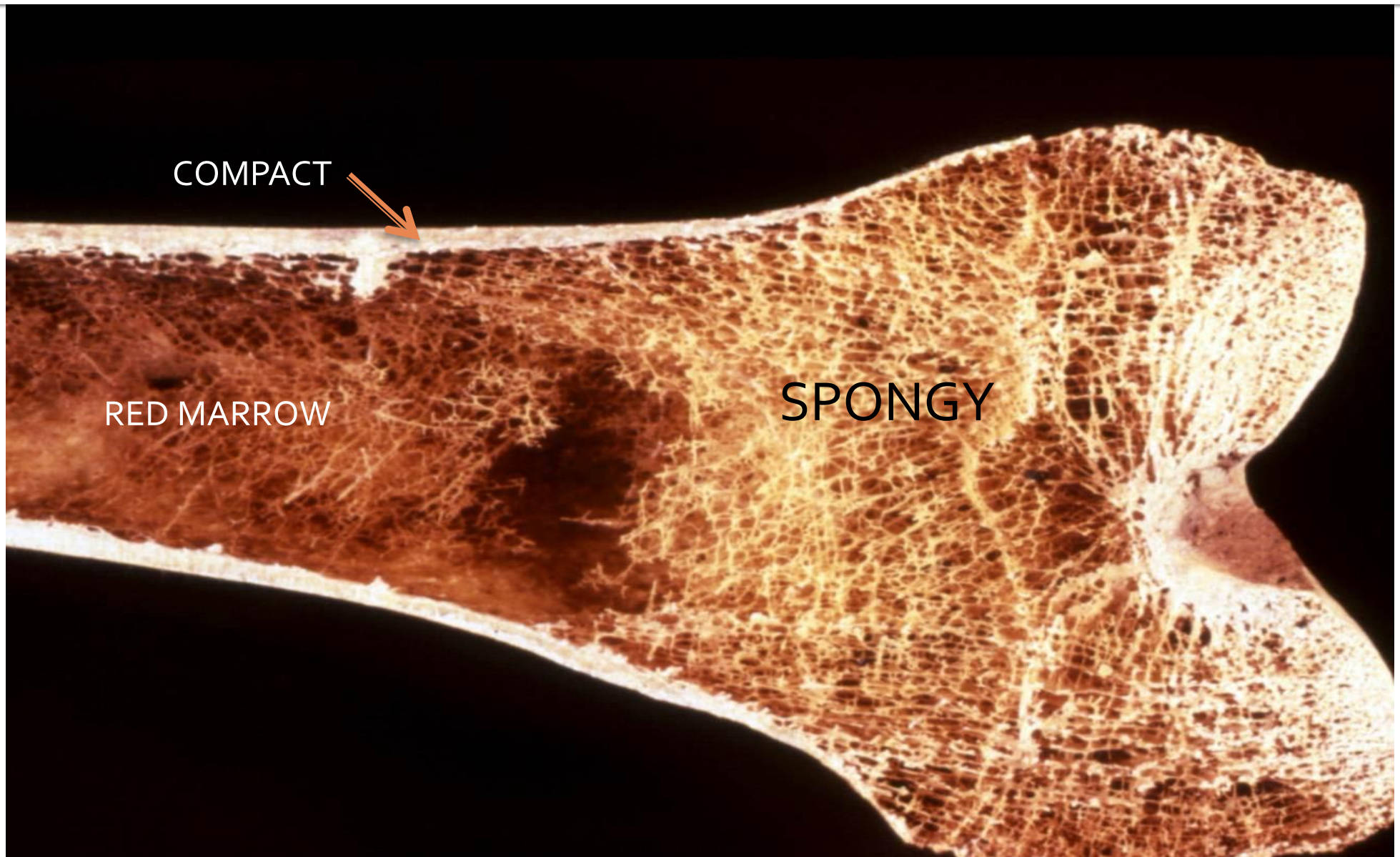
Periosteum – receives sensory (nociceptive) fibers and is vascularized

Fibrous layer contains fibroblasts

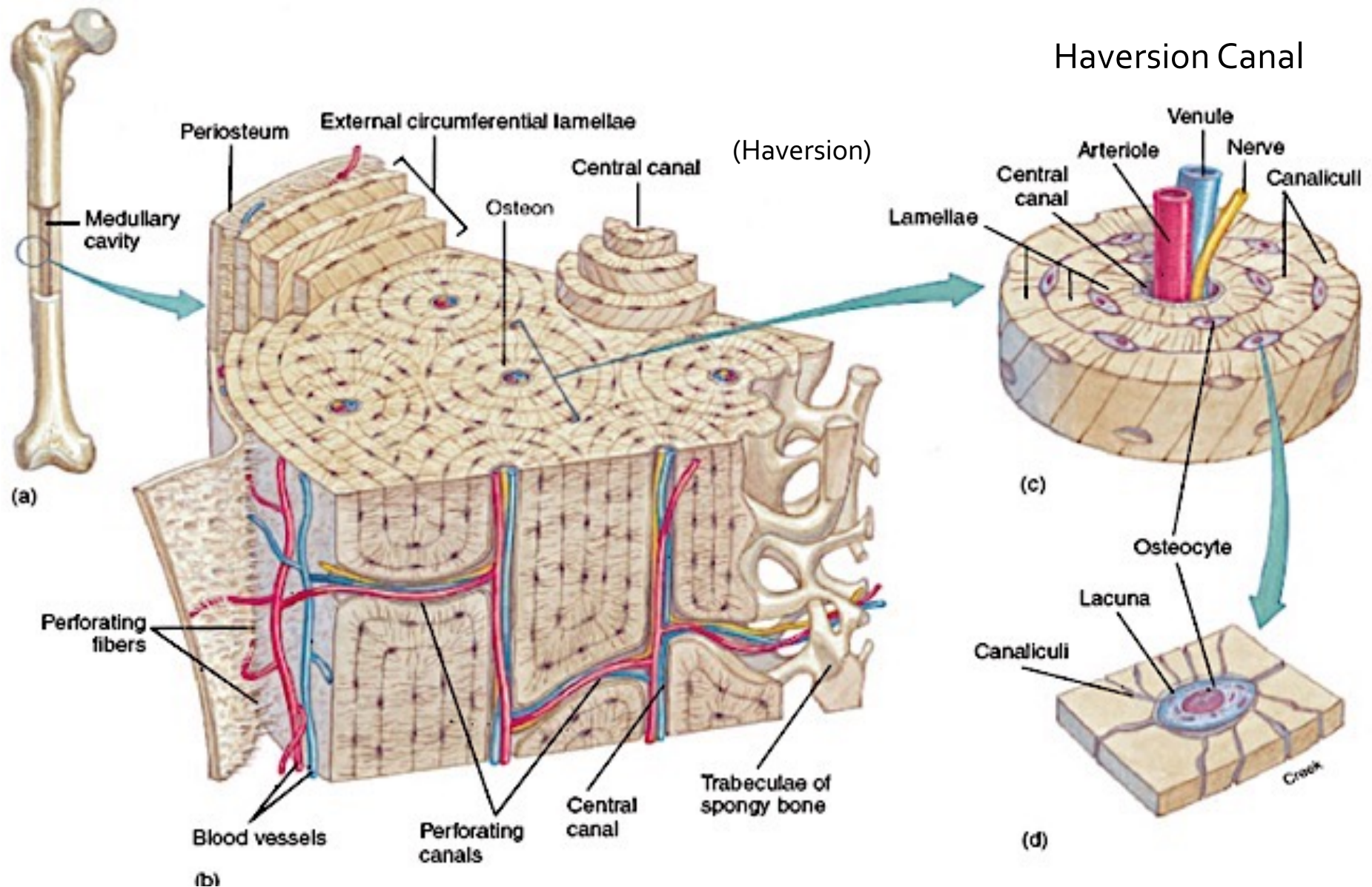
Cellular layer (cambrium) contains progenitor cells of osteoblasts



Section of Long Bone with periosteum removed

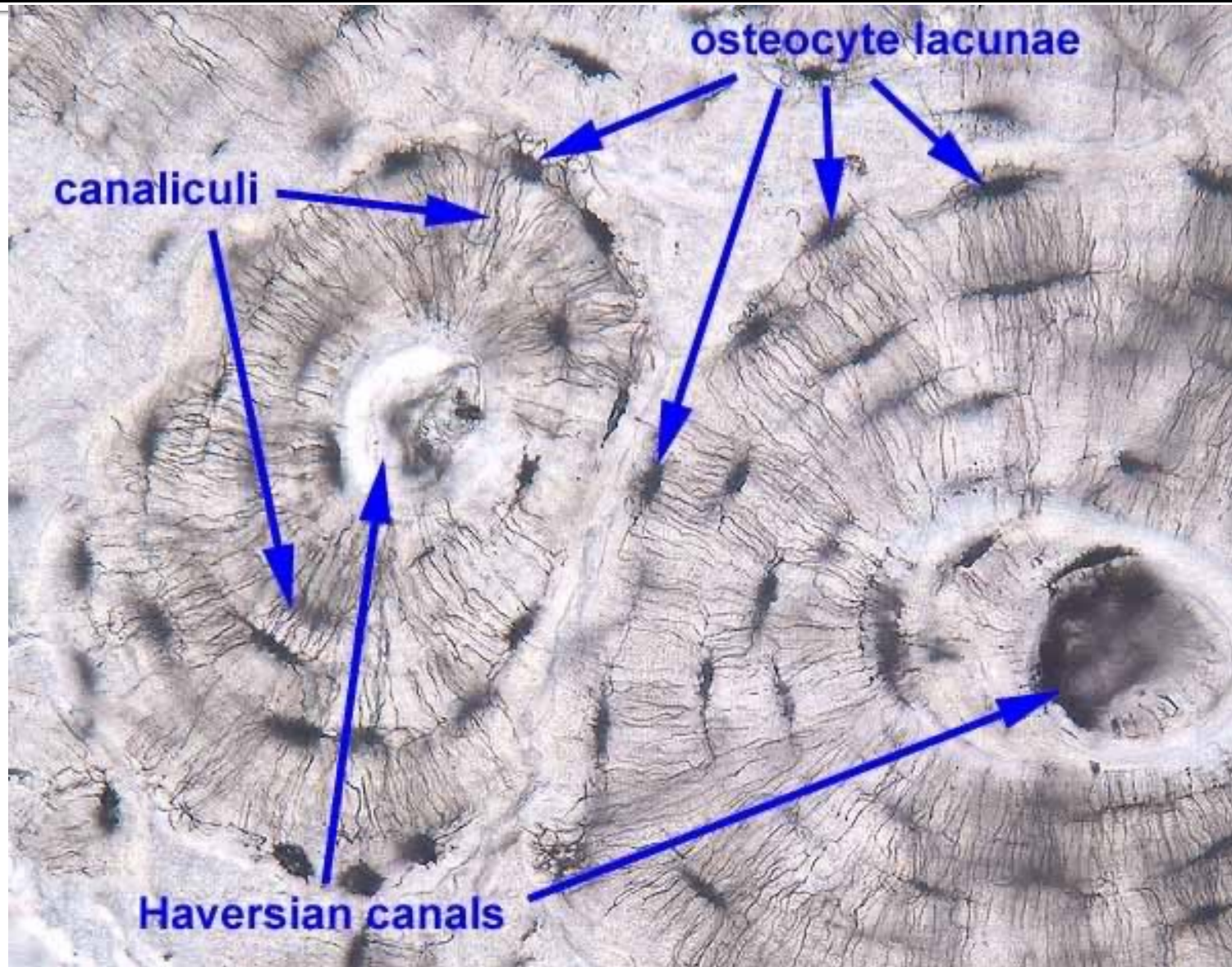


Histology of Osteoid (Compact) Bone



Histology of Osteons

Video on histology of bone: <https://www.youtube.com/watch?v=NGinKPK8wko>



SPONGY (cancellous) bone

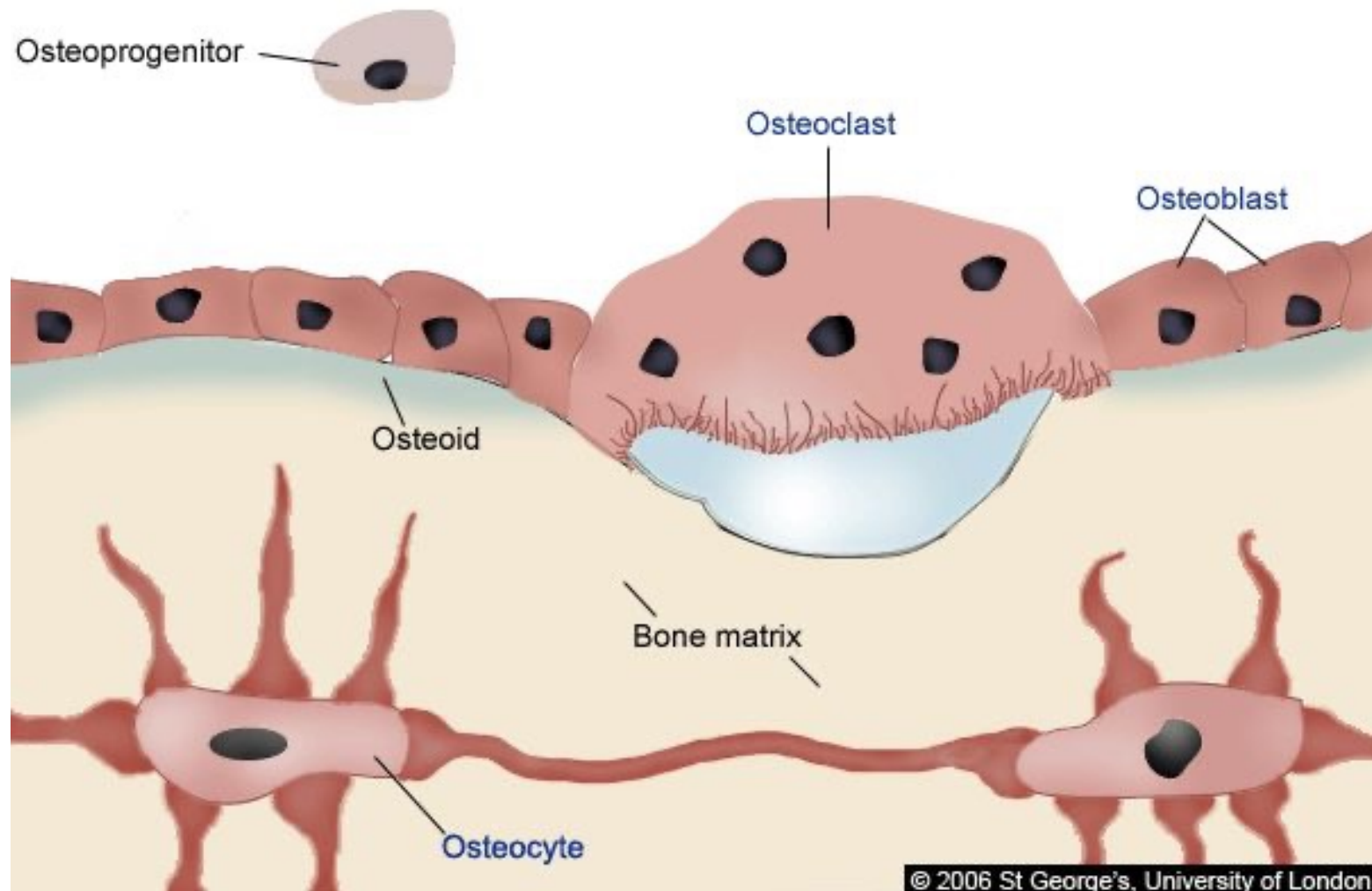


Cells of Bone:

Osteoblasts and Osteocytes

- **OSTEOBLASTS** are the cells that form new bone. They come from the bone marrow. They have a single nucleus. They produce new bone called "osteoid" which is made of bone collagen and other proteins. Then they control calcium and mineral deposition. They are found on the surface of the new bone.
- When a team of osteoblasts has finished filling in a cavity, the cells flatten and line the surface of the bone. These old osteoblasts are also called **LINING CELLS**. They regulate passage of calcium into and out of the bone, and they respond to hormones by making special proteins that activate the osteoclasts.
- **OSTEOCYTES** are cells inside the bone. They also come from osteoblasts. Some of the osteoblasts turn into osteocytes while the new bone is being formed, and the osteocytes then get surrounded by new bone.
- They are not isolated, however, because they send out long branches that connect to the other osteocytes, forming gap junctions.
- These cells can sense pressures or cracks in the bone and help to direct where osteoclasts will dissolve the bone.

Positions of Bone cells



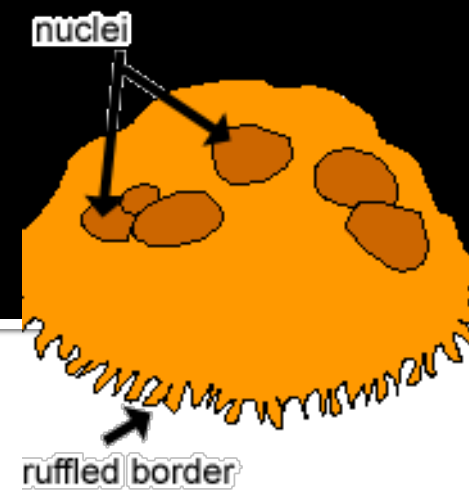
Osteoclasts

Large multinucleated cells, with a 'ruffled border' that resorb bone matrix. They are important for remodeling, growth and repair of bone. (clast - Greek 'to break').

Osteoclasts are not derived from osteoprogenitor cells. They are derived from blood monocytes/macrophages which are derived from hemopoietic cells in the bone marrow. Space around osteoclasts called Howship's lacunae.

Bone re-modelling is necessary for growth: -

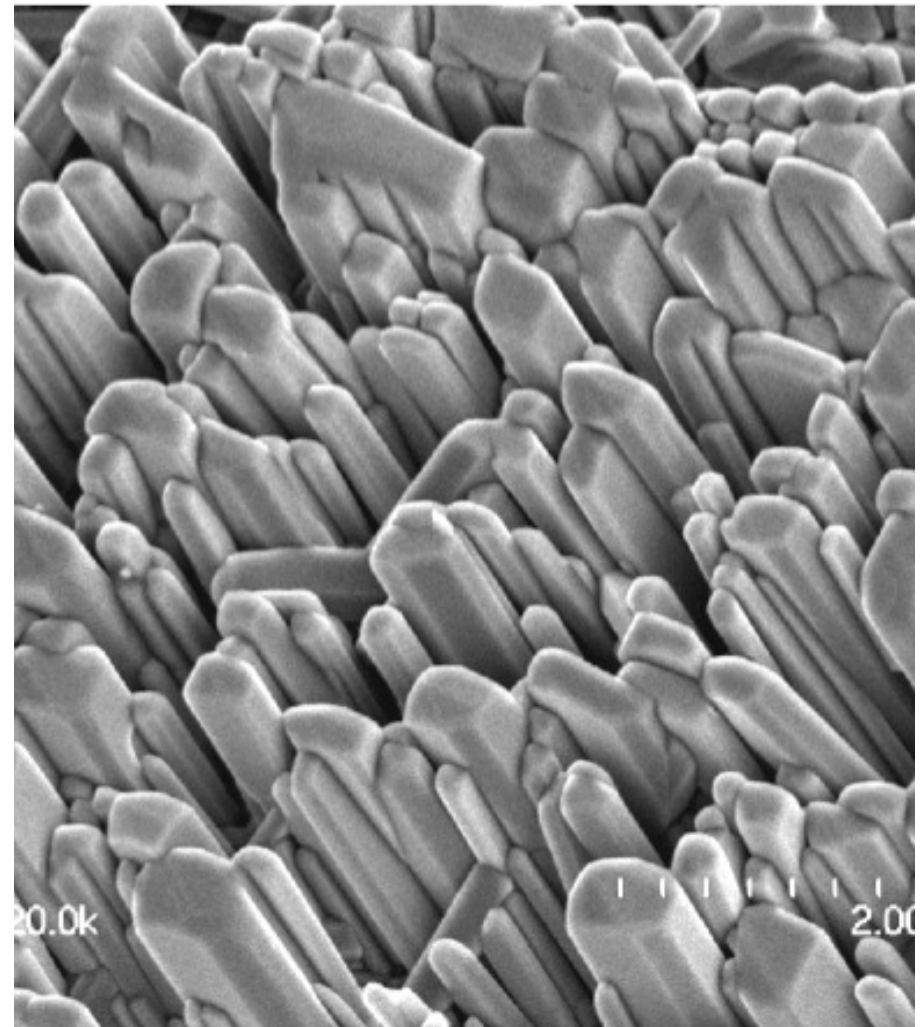
1. Mechanical stresses on the skeleton cause release of calcium, that stimulate bone-remodelling.
2. Hormones also control bone re-modelling. Parathyroid hormone stimulates bone resorption. and calcitonin inhibits resorption.



Extracellular matrix

- Bone matrix is made up of an organic matrix (30%) containing proteoglycans (but less than cartilage), glycosaminoglycans, glycoproteins, osteonectin (anchors bone mineral to collagen) and osteocalcin (calcium binding protein).
- There are collagen fibers (mostly type I (90%), with some type V). Only 25% of bone is water. Almost 70% of bone is made up of bone mineral called hydroxyapatite.
- Before the extracellular matrix is calcified, the tissue is called **osteoid** (bone-like) **tissue**. When the concentrations of calcium and phosphate ions rise high enough, they are deposited into the extracellular matrix, and the bone calcifies.

Hydroxyapatite crystals

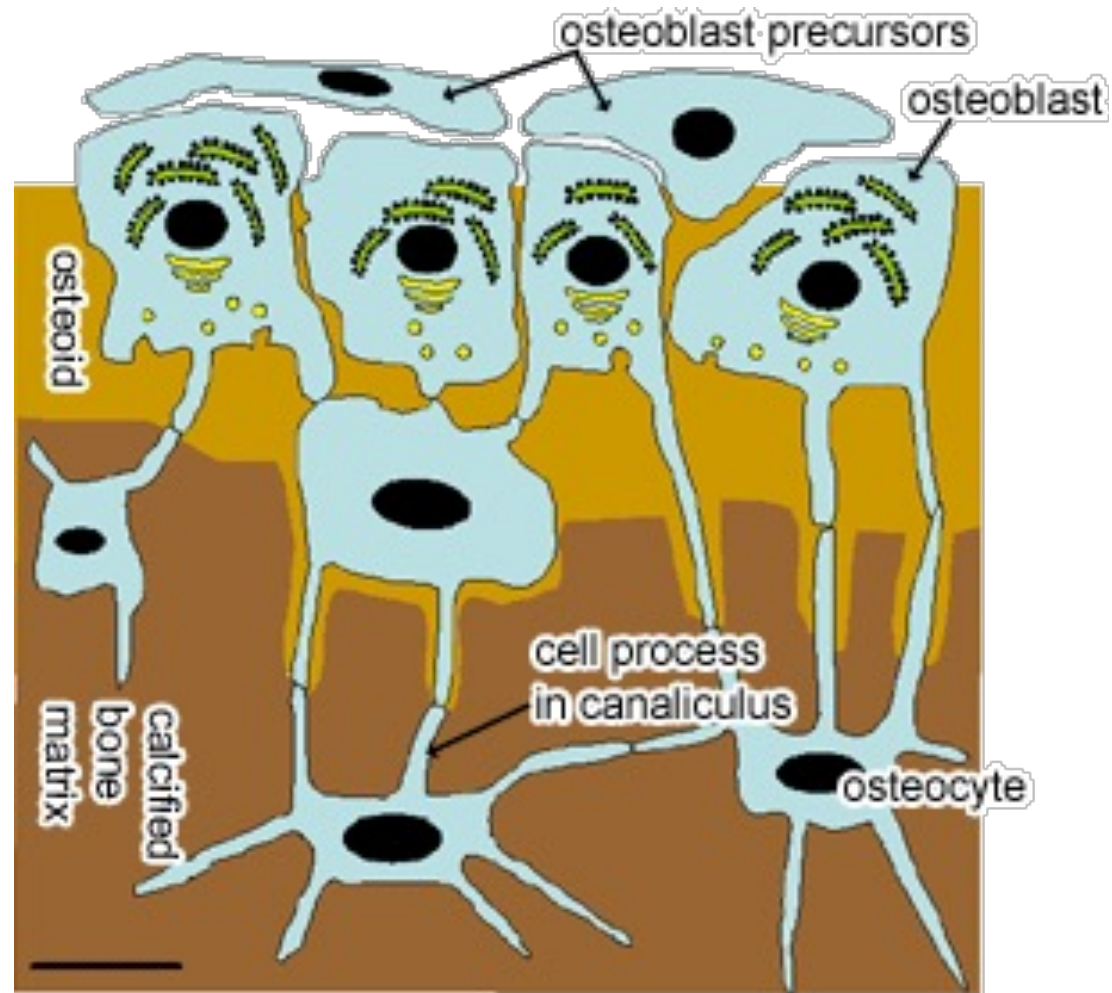


Growth of Bone

- There are two methods of ossification:
- **Endochondral** - formation of bone into a pre-existing cartilage model or scaffold.
- **Intramembranous** - formation of bone directly onto a fibrous connective tissue template, with no cartilage intermediate. This type of ossification occurs in a few specialized places such as the flat bones of skull (i.e. parietal bone, mandible, maxilla and clavicles).

Cells of growing Bone

- **Osteoblasts** line the surface of bone, secrete collagen and the organic matrix of bone (osteoid), which becomes calcified soon after it has been deposited. As they become trapped in the organic matrix, they become **osteocytes**.
- **Osteocytes** maintain bone. Fine processes from these cells ramify through bone, and form gap junctions with other osteocytes.



Video on Bone Formation

- <https://www.youtube.com/watch?v=p-3PuLXp9Wgv=p-3PuLXp9Wg>

Remodeling of Bone

- Note the numerous osteoclasts.
- They produce and release carbonic anhydrase that creates an acidic environment to focally dissolve bone matrix.

